

OpenStax Calculus Volume 2

Section 4.4 Exercises: The Logistic Equation

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For the following problems, consider the logistic equation in the form $P' = CP - P^2$. Draw the directional field and find the stability of the equilibria.

1. $C = 3$
2. $C = 0$
3. $C = -3$
4. Solve the logistic equation for $C = 10$ and an initial condition of $P(0) = 2$.
5. Solve the logistic equation for $C = -10$ and an initial condition of $P(0) = 2$.
6. A population of deer inside a park has a carrying capacity of 200 and a growth rate of 2%. If the initial population is 50 deer, what is the population of deer at any given time?
7. A population of frogs in a pond has a growth rate of 5%. If the initial population is 1000 frogs and the carrying capacity is 6000, what is the population of frogs at any given time?
8. [T] Bacteria grow at a rate of 20% per hour in a petri dish. If there is initially one bacterium and a carrying capacity of 1 million cells, how long does it take to reach 500,000 cells?
9. [T] Rabbits in a park have an initial population of 10 and grow at a rate of 4% per year. If the carrying capacity is 500, at what time does the population reach 100 rabbits?
10. [T] Two monkeys are placed on an island. After 5 years, there are 8 monkeys, and the estimated carrying capacity is 25 monkeys. When does the population of monkeys reach 16 monkeys?
11. [T] A butterfly sanctuary is built that can hold 2000 butterflies, and 400 butterflies are initially moved in. If after 2 months there are now 800 butterflies, when does the population get to 1500 butterflies?

The following problems consider the logistic equation with an added term f for depletion, either through death or emigration:

$$P' = 0.4P \left(1 - \frac{P}{10000} \right) - f.$$

12. [T] The population of trout in a pond is given by $P' = 0.4P \left(1 - \frac{P}{10000} \right) - 400$, where 400 trout are caught per year. Use your calculator or computer software to draw a directional field and draw a few sample solutions. What do you expect for the behavior?

13. In the preceding problem, what are the stabilities of the equilibria $0 < P_1 < P_2$?
14. [T] For the preceding problem, use software to generate a directional field for the value $f = 400$. What are the stabilities of the equilibria?
15. [T] For the preceding problems, use software to generate a directional field for the value $f = 600$. What are the stabilities of the equilibria?
16. [T] For the preceding problems, consider the case where a certain number of fish are added to the pond, or $f = -200$. What are the nonnegative equilibria and their stabilities?

It is more likely that the amount of fishing is governed by the current number of fish present, so instead of a constant number of fish being caught, the rate is proportional to the current number of fish present, with proportionality constant k , as

$$P' = 0.4P(1 - \frac{P}{10000}) - kP.$$

17. [T] For the previous fishing problem, draw a directional field assuming $k = 0.1$. Draw some solutions that exhibit this behavior. What are the equilibria and what are their stabilities?
18. [T] Use software or a calculator to draw directional fields for $k = 0.4$. What are the nonnegative equilibria and their stabilities?
19. [T] Use software or a calculator to draw directional fields for $k = 0.6$. What are the equilibria and their stabilities?
20. Solve this equation, assuming a value of $k = 0.05$ and an initial condition of 2000 fish.
21. Solve this equation, assuming a value of $k = 0.05$ and an initial condition of 5000 fish.

The following problems add in a minimal threshold value for the species to survive, T , which changes the differential equation to $P'(t) = rP(1 - \frac{P}{K})(1 - \frac{T}{P})$.

22. Draw the directional field of the threshold logistic equation, assuming $K = 10$, $r = 0.1$, and $T = 2$. When does the population survive? When does it go extinct?
23. For the preceding problem, solve the logistic threshold equation, assuming the initial condition $P(0) = P_0$.
24. Bengal tigers in a conservation park have a carrying capacity of 100 and need a minimum of 10 to survive. If they grow in population at a rate of 1% per year, with an initial population of 15 tigers, solve for the number of tigers present.
25. A forest containing ring-tailed lemurs in Madagascar has the potential to support 5000 individuals, and the lemur population grows at a rate of 5% per year. A minimum of 500 individuals is needed for the lemurs to survive. Given an initial population of 600 lemurs, solve for the population of lemurs.
26. The population of Florida panthers in the southwestern tip of Florida has an estimated carrying capacity of 250 and grows at a rate of 0.25% per year, and there must be 25 for the population to survive. With an initial population of 30 Florida panthers, how many years will it take to get the Florida panthers off the endangered species list (at least 100)?

The following questions consider the Gompertz equation, a modification for logistic growth, which is often used for modeling cancer growth, specifically the number of tumor cells.

27. The Gompertz equation is given by $P'(t) = \alpha \ln(\frac{K}{P(t)})P(t)$. Draw the directional fields for this equation assuming all parameters are positive, and given that $K = 1$.
28. Assume that for a population, $K = 1000$ and $\alpha = 0.05$. Draw the directional field associated with this differential equation and draw a few solutions. What is the behavior of the population?
29. Solve the Gompertz equation for generic α and K and $P(0) = P_0$.
30. [T] The Gompertz equation has been used to model tumor growth in the human body. Starting from one tumor cell on day 1 and assuming $\alpha = 0.1$ and a carrying capacity of 10 million cells, how long does it take to reach “detection” stage at 5 million cells?
31. [T] It is estimated that the world human population reached 3 billion people in 1959 and 6 billion in 1999. Assuming a carrying capacity of 16 billion humans, write and solve the differential equation for logistic growth, and determine what year the population reached 7 billion.
32. [T] It is estimated that the world human population reached 3 billion people in 1959 and 6 billion in 1999. Assuming a carrying capacity of 16 billion humans, write and solve the differential equation for Gompertz growth, and determine what year the population reached 7 billion. Was logistic growth or Gompertz growth more accurate, considering world population reached 7 billion on October 31, 2011?
33. Show that the population grows fastest when it reaches half the carrying capacity for the logistic equation $P' = rP(1 - \frac{P}{K})$.
34. When does population increase the fastest in the threshold logistic equation $P'(t) = rP(1 - \frac{P}{K})(1 - \frac{T}{P})$?
35. When does population increase the fastest for the Gompertz equation $P'(t) = \alpha \ln(\frac{K}{P(t)})P(t)$?

Below is a table of the populations of whooping cranes in the wild from 1940 to 2000. The population rebounded from near extinction after conservation efforts began. The following problems consider applying population models to fit the data. Assume a carrying capacity of 10,000 cranes. Fit the data assuming years since 1940 (so your initial population at time 0 would be 22 cranes).

36. Find the equation and parameter r that best fit the data for the logistic equation.
37. Find the equation and parameters r and T that best fit the data for the threshold logistic equation.
38. Find the equation and parameter α that best fit the data for the Gompertz equation.
39. Graph all three solutions and the data on the same graph. Which model appears to be most accurate?
40. Using the three equations found in the previous problems, estimate the population in 2010 (year 70 after conservation). The real population measured at that time was 437. Which model is most accurate?